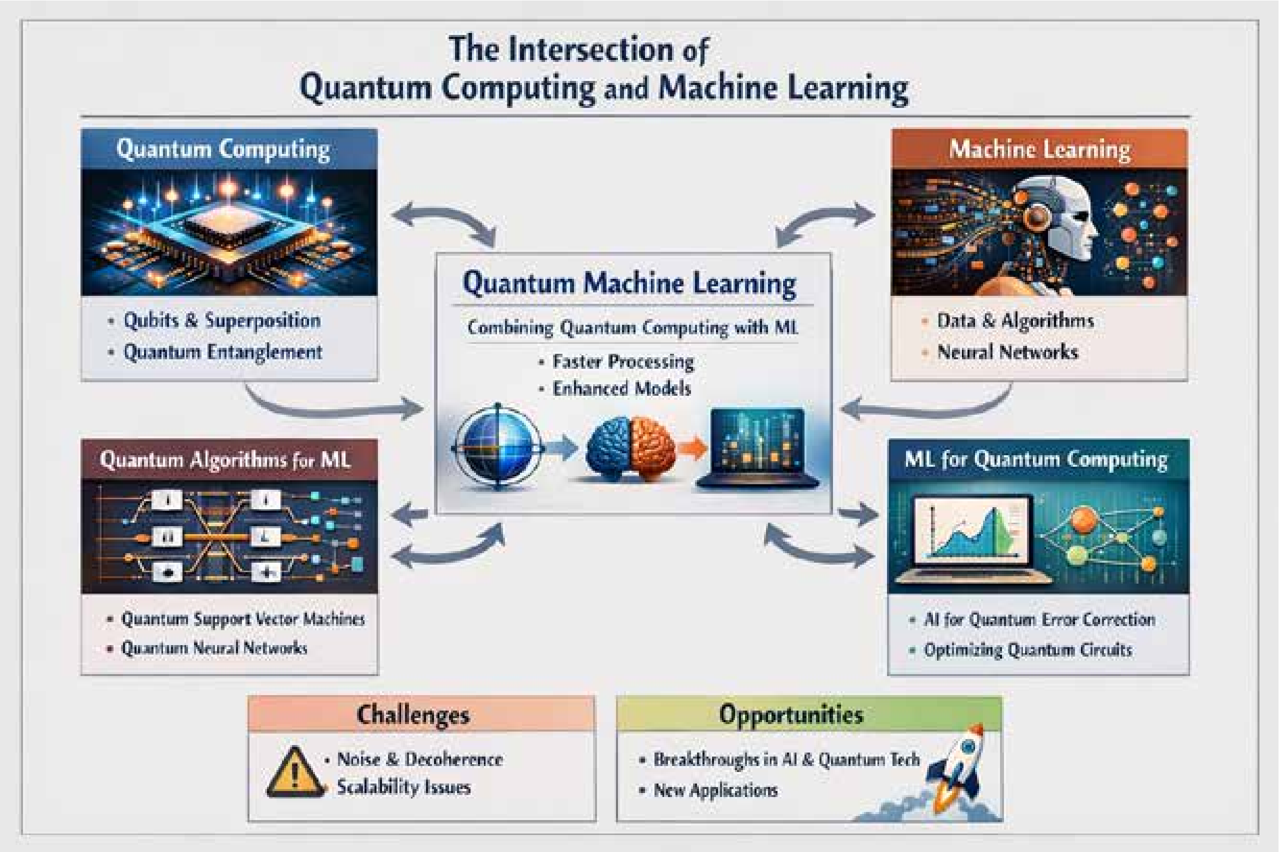


Figure 1: Quantum computing and machine learning

advancements are not merely about speed; they also enable the tackling of previously unsolvable problems, especially in fields like optimisation and pattern recognition. As quantum computers become more accessible, AI practitioners will increasingly leverage these algorithms to gain a competitive edge, leading to faster innovation cycles and deeper analytical capabilities.

## Quantum neural networks: Fact or fiction?

The concept of quantum neural networks (QNNs) sparks much debate within the scientific community. While the idea of combining neural networks with quantum computing is captivating, practical implementations remain in their infancy. Researchers are actively exploring how quantum properties might enhance learning processes, but most QNNs today are experimental rather than fully operational. Despite this, theoretical models suggest that quantum neural networks could offer significant improvements in how AI learns and generalises. The journey from fiction to fact is ongoing, and each breakthrough brings us closer to real-world applications.

## Early quantum AI success stories

Several pioneering organisations have begun experimenting with quantum machine learning, yielding promising results. For instance, pharmaceutical companies are using quantum AI to accelerate drug discovery by analysing molecular structures at a level unattainable by classical computers. In finance, quantum algorithms are being tested for portfolio optimisation and risk assessment, offering more precise forecasts. Even sectors like logistics and supply chain management are witnessing early successes, where quantum-enhanced models streamline operations and enhance decision-making. These success stories underscore the transformative potential of quantum AI across industries.

## Quantum AI in BFSI

When people in banking, financial services, and insurance (BFSI) talk about quantum AI, the conversation usually starts with one simple question: where would it help first? In BFSI, the pressure points are consistent—risk, fraud, pricing, and complex optimisation problems that classical systems can solve, but often only with trade-offs in speed or accuracy. Quantum-enhanced machine learning offers a different way to search huge solution spaces and spot patterns in noisy, high-dimensional data.